Geometric sequences

1 By choosing any term after the first and dividing it by the previous term.

2

a -8 **b** -0.9 **c** $\frac{1}{4}$

3

a $108, 324$ **b** $-768, 3072$ **c** $\frac{27}{64}, \frac{81}{256}$ **d** $\frac{81}{4}, -\frac{243}{8}$

4

a

i 2 **ii** 2 **iii** 2

b $T_5 = -128$

5

a $6, 24, 96, 384$

b $7, -14, 28, -56$

c $-2, -6, -18, -54$

d $1.3, -5.2, 20.8, -83.2$

e $700\,000, 728\,000, 757\,120, 787\,404.8$

6 Neither

7

a

i Geometric as terms are formed by multiplying by a constant ratio.

ii $r = -9$

b

i Arithmetic as terms are generated by adding a constant.

ii $d = 4$

8

a $x = 20, y = -1280$

b $a = 3, b = -\frac{3}{5}, c = \frac{3}{625}$

9 Yes, as $\frac{t_3}{t_1} = r^2$ and $\frac{t_5}{t_3} = r^2$ so there is a common ratio of r^2.

10 $T_{10} = \frac{1}{729}$

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11 $T_4 = -150$

12 $T_3 = 501\,760$

13 8

14

a $r = \dfrac{2}{3}$ **b** $9, 6, 4$

15 $18, 12, 8, \dfrac{16}{3}, \dfrac{32}{9}$

16 $\dfrac{5}{3}, 5, 15$

17

a $\sqrt{2}$ **b** $16\sqrt{2}$ **c** $n = 13$

Explicit rules

18

a

i $3, 12, 48, 192$ **ii** $r = 4$

b

i $-4, 12, -36, 108$ **ii** $r = -3$

19

a $r = \pm 2$ **b** $a = 4$ **c** $T_n = 4 \times 2^{n-1}$

20

a

n	1	2	3	4	10
T_n	25	5	1	$\dfrac{1}{5}$	$\dfrac{1}{78\,125}$

b $r = \dfrac{1}{5}$

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21

a

n	1	2	3	4	6
T_n	-72	96	-128	$\frac{512}{3}$	$\frac{8192}{27}$

b $r = -\frac{4}{3}$

22

a

i $r = 8$

ii $T_n = 5 \times 8^{n-1}$

iii $T_{10} = 671\,088\,640$

b

i $r = -3$

ii $T_n = 7 \times (-3)^{n-1}$

iii $T_{12} = -1\,240\,029$

c

i $r = 3$

ii $T_n = -5 \times 3^{n-1}$

iii $T_{11} = -295\,245$

d

i $r = \frac{8}{3}$

ii $T_n = -2 \times \left(\frac{8}{3}\right)^{n-1}$

iii $T_7 = -\frac{8192}{81}$

23

a

n	1	2	3	4	5
T_n	5	-20	80	-320	1280

b

n	1	2	3	4	5
T_n	-27	-36	-48	-64	$-\frac{256}{3}$

24

a $r = -4$

b $T_n = -9 \times (-4)^{n-1}$

25

a $2\,(-8)^{n-1}$

b $8192, -65\,536, 524\,288$

c $2\,147\,483\,648$

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26

a $-0.3 \times 5^{n-1}$

b $-187.5, -937.5, -4687.5$

Recursive rules

27 $T_n = \frac{1}{5}T_{n-1}, T_1 = 9000$

28

a $r = 4$

b $T_{n+1} = 4T_n, T_1 = 6$

29

a $r = \pm 4$

b $T_n = 4T_{n-1}, T_1 = 5$

c $T_n = -4T_{n-1}, T_1 = 5$

30

a $r = 2$

b $a = 4$

c $T_{n+1} = 2T_n, T_1 = 4$

31

a $r = \pm\frac{1}{5}$

b $62\,500$

c $T_n = \frac{1}{5} \times T_{n-1}, T_1 = 62\,500$

d $T_n = -\frac{1}{5} \times T_{n-1}, T_1 = 62\,500$

32

a -343

b -2401

c $-16\,807$

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Sequences on the coordinate plane

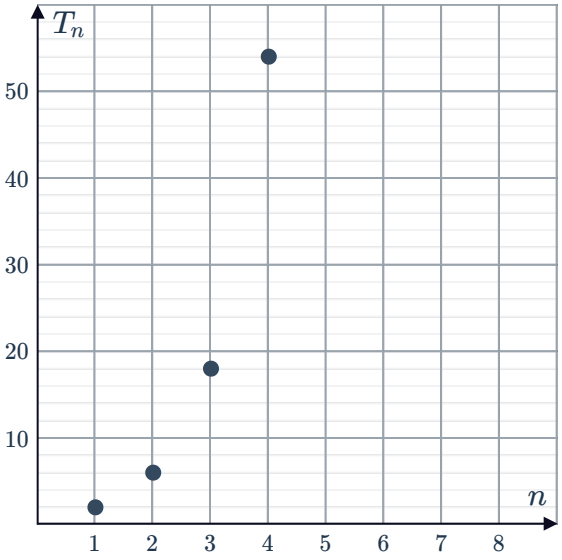
33

a

n	1	2	3	4	10
T_n	2	6	18	54	39 366

b 3

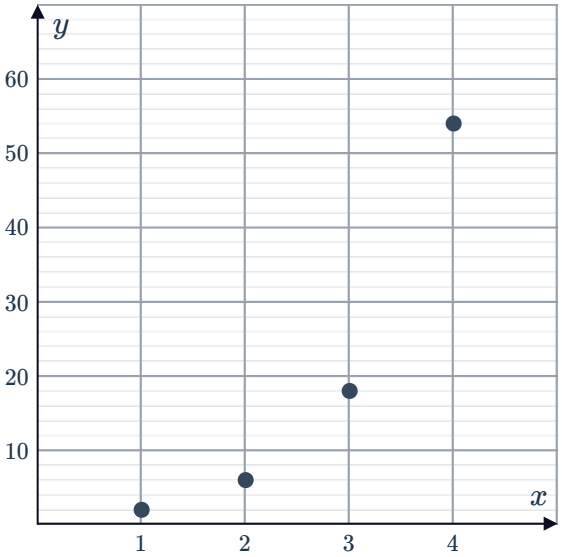
c



d Curved line

34

a



b Geometric

c $T_n = 3T_{n-1}, T_1 = 2$

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35

a

n	1	2	3	4
T_n	3	6	12	24

b $r = 2$ **c** $T_n = 3 \times 2^{n-1}$ **d** $T_{10} = 1536$

36

a

n	1	2	3	4
T_n	2	-6	18	-54

b $r = -3$

c $T_4 = -54$ **d** $T_n = 2 \times (-3)^{n-1}$

e $T_{10} = -39\,366$

37

a

n	1	2	3
T_n	-2	-6	-18

b $r = 3$ **c** $T_n = -2 \times 3^{n-1}$

38

a -2 **b** $r = 3$ **c** $T_n = -2 \times 3^{n-1}$

39

a $r = 2$ **b** $T_n = 3 \times 2^{n-1}$ **c** $T_k = -3 \times 2^{k-1}$

40 $T_1 = 5, T_n = 3T_{n-1}$

41

a $1, 2, 4, 8, 16$

b Geometric, because each term increases from the last by a constant factor.

c $T_{n+1} = 2T_n, T_1 = 1$

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Applications

42

a 8.48 cm **b** 10.10 cm

c $H_n = 1.06H_{n-1}, H_0 = 8$

43

a $\$77\,400$ **b** $\$57\,245.04$

c $V_n = 0.86V_{n-1}, V_0 = 90\,000$

44

a $\$3.74$ **b** $\$3.94$

c $V_n = 1.026V_{n-1}, V_0 = 3.65$

45

a $\$65\,536$ **b** $\$268\,435\,456$ **c** $\$8191$ **d** $\$536\,870\,911$

46

a 759 nurses **b** 3980 nurses

47

a 12% **b** $19\,994$

48

a $1:1$ **b** $20\,736\text{ in}^2$ **c** $n = 2$

49

a 6832 yd^3 **b** 7168 yd^3

50

a 1216 bacteria **b** 9 bacteria

51

a $1\,140\,152.14\text{ m}^3$ **b** $1\,370\,000\text{ m}^3$

52

a 1.2 **b** 51.84 mm

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53 $n = 5$

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